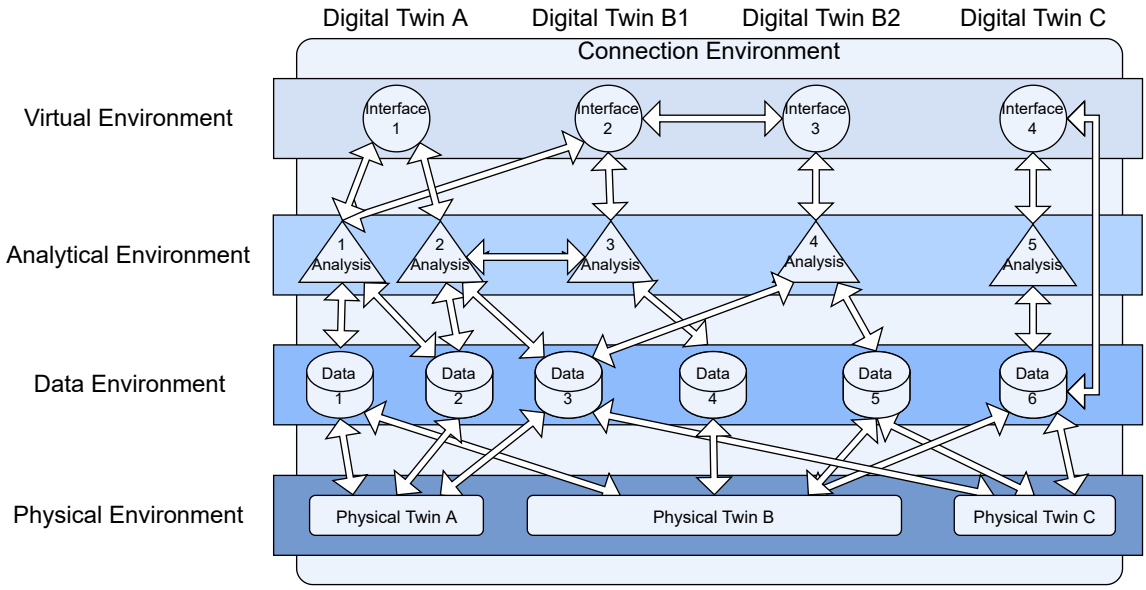

Figure 10. Composition of Components to form Digital Twins. Each DT is represented as a column. The Connection Environment (represented by arrows) connects instances in all environments across DTs. The other environments are aligned in rows to show the implementation across DTs. Instances are shown in each environment as rectangles, cylinders, triangles or circles. Data from the Data Environment can processed in the Analytical Environment or displayed directly in the Virtual Environment. DTs can also be siloed (e.g., DT C) but more often will use the Connection Environment with complex dependencies across all environments. For example, DT A uses Data 3 and Analysis 3 from PT B. PT B has two DTs where DT B2 is composed into DT B1 in the Virtual Environment by combining interface 2 and 3. DT B1 & B2 are using data that is collected from all PTs. Despite DT C being siloed, PT C is evaluated in DT A through Data 3 and Analysis 2. Note that this abstraction makes no claims about the mode of data collection in the Physical Environment that may be with either local sensors or remote sensing. Generated with http://draw.io.

composability of DTs suggests that, by extension, these applications should be composable, too. For example, DTs in industry, health care, and transportation could exchange vital information to improve their effectiveness, but, to the best of our knowledge, this has not been accomplished in practice. To understand how DTs of these varied systems could be composed in the context of truly smart cities, we categorise each application by its system type and location type to form a taxonomy of DT contexts (see Table 1).

Table 1. Taxonomy of Digital Twin Context Types

		System Type	
		Open	Closed
Location Type	Stationary	Cities, Economies & Countries	Industrial processes
	Mobile	End users, goods & vehicles	Body (health) & machines (internal)

The system type of a DT is either open or closed [202]. Open systems are characterised by an exchange with the wider environment in which they are embedded, whereas closed system do not directly exchange with the environment and are usually observed in isolation. In addition, closed systems usually supervise well-defined processes, whereas open systems try to characterise ill-defined processes. Obviously, no system is entirely isolated, but industrial processes or health applications can be considered relatively closed systems with usually well-defined interactions with the environment. In contrast, cities and traffic [177] have no clear limit to what is entering from the surrounding environment, and congestion may be a side effect of this ill-defined boundary.

The location type of a DT describes the relation between the unit of observation and the sensors as active or passive sensing [203]. In stationary systems, the sensors are fixed in the environment and observe throughput. In mobile systems, the sensors are moving around either with the observed unit or independent of it. They may also capture environmental conditions around the observed unit such as in human navigation [204] and health monitoring [205].

When composing different types of DTs, it is important to understand their characteristics. Mobile DTs may only temporarily compose with Stationary DTs, while Closed DTs often form units within Open DTs. Open Stationary DTs often form the overarching type of DT that contains other types of DTs, whereas Mobile Closed DTs usually describe atomic units in composed DTs. Typically, Mobile Closed DTs can be part of Mobile Open DTs when a person has a Body Area Network (BAN) [206] for health care applications and a Personal Area Network (PAN) [207] to interact with the environment. In this example, the PAN will provide a wide range of interactions that can be composed with the Open Stationary DT of a Smart City, whereas the BAN will only share limited information. Similarly, a Closed Stationary DT for a Smart Factory within Open Stationary DT of a Smart City will only share limited information.

4.2. Servicisation of Digital Twins

While the word DT suggests some form of unitary entity, it would be more accurate to understand DTs as context-aware representations of Things in the cloud composed of services that act upon the abstraction. Services can range from interacting with the environment to representing the sensed and form the bedrock of ubiquitous computing [46]. Basic services again can be composed into higher-order services to address more complex issues.

With digital representation by the DT, it is possible to turn everything into a service—or *servicise* [208–211]. In other words, the customer buys the temporary use of a service associated with a product instead of the product itself. The cost of developing and maintaining a DT's infrastructure justifies this approach because it is more efficient to reuse resources rather than

to duplicate functionalities [211]. Therefore, DTs can be freely composed of services that conform to the requirements of the Connection Environment .

The three most common themes in "as a Service" (*aaS) are Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS). Newer types of services can be subsumed in these categories, including Sensing as a Services (SenaaS) [212,213], Semantics as a Service (SemaaS) [175], and Robots as a Service (RaaS) [214]. Notably, Infrastructure, Platform, and Software "as a Service" form an interesting interaction with the DT components because they can provide abstractions for particular components (see Figure 2). From an abstraction perspective, each replaces the lower layers of the stack with an interface that hides implementation complexity. In this section, we will use the example of weather services as part of a smart city to explain how IaaS, PaaS, and SaaS can map to the components of a DT (see Figure 11).

4.2.1. Infrastructure as a Service (IaaS)

A DT requires the creation of a Physical Environment as the representation of the PT. The Physical Environment provides access to and measurement of the PT using infrastructure to collect raw data (see Figure 2). Creating and maintaining the DT infrastructure is the most physically laborious part of developing a DT. This severely limits the number of entities that can afford to create a DT on a city scale. However, provided as a service, it may be possible to finance the required infrastructure in aggregate and reuse it across multiple DTs. When creating a DT, the IaaS approach allows for the reuse of a service provider's communication network, sensor network, and remote sensing units rather than having to build one's own system from scratch. Therefore, IaaS enables to have alternative competing and composable DTs based on the same or different data source(s). There are also more specialised intrastructure services that are available such as Sensing as a Services (SenaaS) [212,213], Remote Sensing as a Service (RSaaS) [215], and Robots as a Service (RaaS) [214].

In our weather service example (see Figure 11), the sensors, network, and remote sensing units in a smart city could be servicised. The city government or a private provider could implement a policy to maintain such an IaaS to make digital services in a city more accessible, enabling the creation of different city DTs. This would allow new entrants to easily develop their own DTs that provide large-scale digital services without having to build physical hardware with redundant functionality [209].

At the same time, IaaS may raise some concerns regarding security. The IoT used for IaaS consists of billions of locatable, addressable, and readable devices [190]. There are three avenues by which security in IoT must be enforced: information, physical infrastructure, and management practices [188]. Security breaches from unexpected behaviour may arise from data collection and management via privacy protection problems, heterogeneous network authentication, and access control problems [188–190,216]. The fundamental challenges for all of these topics are questions of trust: how to give whom access for what purpose and for what duration. Over the last decade, the IEEE and IETF have laid the foundations to build a security infrastructure along the IEEE 802.15.4 standard [217,218] and the IETF LLN stack [207]. These efforts include sensor network security [183], wireless sensor network security [182], and dedicated secure protocols such as CoAP/Lithe [219,220], 6LowPSec [221] and RPL [222]. Ultimately, such as with the Internet, security will be an ongoing process that must be attended at all times.

4.2.2. Platform as a Service (PaaS)

PaaS extends access to hardware by simplifying interfaces. PaaS usually packages digital representations into a Data Environment and thus reduces the complexity of accessing the data. Platforms vary largely and are usually tied to one of the five IoT application areas

While the term IoT has recently become the most common term for the linkage between the virtual and the real, it was not always the case. Previously, a set of competing ideas (e.g., CPS, WSAN, and Ubiquitous Computing) co-created the ecosystem that today is generously labelled IoT and which laid the hardware foundation for DTs. These ideas were closely related but ultimately based on different assumptions because they originated from different fields. According to public perception, the terminology has settled on IoT even as the underlying technology is still quickly diversifying and expanding. With current trends, DTs appear to be a concept that is set to overtake the IoT in terms of public perception (see Figure 12).

Another issue is that IoT is a buzzword that is difficult to grasp because it is overloaded with meaning. Simply tapping into the literature is overwhelming and, at the same time, highly specific. The many technologies peddled under the umbrella-term IoT would be more distinctive if labelled specifically according to their function. While the technological umbrella-term is now well-established, if ill-defined, there is still a lingering sense that different fields of application are uniquely distinct and require special attention. Research and development is thus heavily directed towards five distinct real-world applications that promise the biggest gains: industrial application, urban management, health care, vehicular applications, and end-user applications [47,164,201]. This ingrained belief regarding the distinction leads to difficulties in synthesising common knowledge across applications. Overcoming this barrier could help the interdisciplinary fields of IoT and DTs in many ways. Not only would it enable learning from each other to clarify their theoretical backgrounds, but it would also make the use of IoT as a basis for DTs more accessible. Indeed, a clearly communicated and unified approach to IoT would lower the barriers of entry tremendously.

Weiser [46] postulated the goal of Ubiquitous Computing to make the computations disappear in the mundane of the environment. To obtain services that hide in plain sight, three categories of design considerations have been brought forward that need to be considered for IoT to work: fundamental properties of a Thing, network communication properties, and real-time interaction properties. The fundamental properties of a Thing include existence, sense of self, connectivity, interactivity, dynamicity and sometimes environmental awareness [177,190,238]. Much research has also been conducted on properties to make network communication more efficient, including energy, latency, throughput, modular design for improved scalability, topology, security of information, safety [42,177,238], deployment, mobility, cost, efficient resource usage, heterogeneity, communication modality, infrastructure, coverage, network size, and lifetime [177,238]. Real-time requirements in industrial and healthcare settings have brought forth another set of properties that need to be considered to enable real-time interaction with IoT networks: predictable delays and latencies, congestion free communication, accurate time synchronisation, real-time assurance, interoperability with existing infrastructure, and prioritised communication [177].

The historic development IoT and the evolution of definitions has been addressed in detail by Atzori *et al.* [231]. The work identified three generations of IoT development with different foci that have resulted in the current diffusive definitions (see Table 3). An additional new generation is identified in this review paper that exemplifies the most current trends of DT research in IoT that was not reported in [231]. Notably, while these four generations focus on distinctive features, they are effectively contemporary because these generations are still actively researched. To make sense of these concurrent generations, it is best to understand each generation as a necessary condition to start work on the following generations.

The first generation is driven by the rise of RFID tags [264] to digitally distinguish Things. At this time, IoT was often taken to be synonymous with RFID. This generation is tightly coupled to Ubiquitous Computing [46], Machine-to-Machine (M2M), and CPSs as the RFID technologies ultimately derive from these research fields. However, by the late 2000s, most of these technologies had been tagged IoT. The second generation is driven by

Table 3. Waves of IoT generations

	Generation I	Generation II	Generation III	Generation IV
Earliest Mention	1999	2010	2011	2012
Technologies	RFID	Inter-networking	Cloud computing	Fog Computing
	M2M	Web of Things	ICN	DTs
	Object integrations	Constrained devices	Semantic IoT	Opportunistic IoT
Focus	Digitalisation	Networking	Centralisation	Decentralisation
Source	[231,264]	[231,281]	[231,267]	own definition, [7,282,283]

interconnecting Things instead of just centralising the information flow. It focuses on the network component of IoT devices and realistically brings the *Internet* into the IoT [281]. Control of information becomes more central as well, reinforcing the connections to its roots in CPSs [135]. The third generation is driven by a re-centralisation around cloud technologies that matured meanwhile [267,279,280]. This generation focuses on processing data and generating knowledge. The fourth generation is driven by a second wave of decentralisation due to time-sensitivity of applications, fog computing [282], and the rise of DTs [7] and servicisation [211]. As questions of processing have been addressed, the newest generation brings more efficiency to IoT. IoT is often mentioned as a multiplier that makes other technologies more attractive [231], contributing to the diffusion of the term.

4.3.6. M2M

> Machine-to-Machine (M2M) communication allows Things and network components to interact automatically with each other without human actions, thus enabling automated formation of CPSs or DTs on different scales.

The complexity of an IoT system underpinning a DT requires much communication between each involved Thing and the network infrastructure to support a control system [135,233,284]. These communication protocols have been aggregated and named M2M communication. M2M is an ongoing research field and is usually required for all other IoT-related and DT-related activities [285–288]. Essentially, M2M describes how IoT components interact with each other and can be understood as part of the implementation of the Connection Environment of a DT. As such, M2M, IoT, and DTs cannot be meaningfully separated, although they do not constitute the same concept. Before IoT became the defining word for ICT, M2M was a contender for being the umbrella-term for these technologies.

The number of devices connected to IoT today surpasses the number of devices manually connected to the Internet. Thus, it is impossible to manually manage the communication between devices by connecting them or maintaining them [177,289–291]. This aspect triggered the development of the M2M concept in the early 2000s [287,292] followed by standardisations such as oneM2M [152] in the 2010s. At the centre of M2M activities are the protocols that are used to exchange information between devices. Protocols describe predefined procedures of how to initiate, maintain, and reestablish communication [152]. While some protocols are high-level, they are often bound by the physical limitations of the communication capabilities of the constrained devices. The complexity of the classical standard Internet Protocol (IP) was quickly disqualified as a platform [293] because of the assumptions that the computing infrastructure was powerful, wired, and connected and that communication was a negligible overhead. However, constrained devices have been highly optimised in terms of energy consumption, size, and costs such that communication has become the major bottleneck [294].

The physical infrastructure to connect constrained devices to the network is usually referred to as physical / PHY [180], and the data link / MAC address is used to uniquely identify single units [295–297]. These terms originate from the OSI model and refer to the physical and data link layers between devices [298]. Originally, a link would be a copper or fibre cable. However, wireless transmission has adopted the same terminology, so PHY usually refers to some kind of antenna that uses a pattern of signals to encode information [250]. To enable a more efficient use of bandwidth and potentially allow for more devices, Cognitive Radios allow secondary users to use communication channels while primary users are idle [286,299,300]. The main trade-off for the PHY connection is between data rates, connection distances, and energy consumption. Currently, technology can be approximately grouped into four application areas: Body Area Networks (BAN), Personal Area Networks (PAN), Local Area Networks (LAN), and Wide Area Networks (WAN) (see Table 4 for an overview).

Table 4. Communication technologies for PHY layer

Scope	Abbreviation	Technologies	Reviews
Body Area	BAN	BLE, Zigbee	[206,301–304]
Personal Area	PAN	UWB, Bluetooth, WiFi	[305,306]
Local Area	LAN	WiFi, LTE	[306,307]
Wide Area	WAN	LoRaWAN, Sigfox, NB-IoT	[1,308–313]

The large standardising bodies, IEEE and IETF, approach protocols in constrained environments based on expansion of the IEEE Standard 802.15 [314,315] and the IETF LLN protocol [293,294] to handle IP. In addition, there are technological niches that have not been fully covered by those standardising organisations such as the Low Power Wide Area Networks (LPWAN) where protocols vary and are highly incompatible, representing a low level of industry cooperation [308–311]. However, several consortia have developed around the different core technologies associated with each protocol, and only subsequent years will show how prevalent each WAN technology has become.

The commonality among all protocols is that they are simplified to move data from the constrained devices to some kind of translation unit that transfers the data into the IP protocol, making it widely available on the Internet [316–318]. This translation is necessary to cope with the reduced computing capacity of constrained devices as well as questions of how to transmit data and scale local IoT networks for use in a DT. A difficulty arises for these protocols because the translation unit of constrained devices do not only send data (such as with sensors) but may also receive data (such as with actuators). This aspect requires a bidirectional translation protocol that can uniquely address every node from the outside [319].

4.3.7. Computation-on-the-wire / Dew / Mist / Fog / Cloud and Edge Computing

Computation-on-the-wire computes all necessary transformations on data at the best possible point in the network according to the *subsidiarity principle* to deliver Ubiquitous Computing for CPSs such as DTs.

The trend to locally compute, distribute, and store data in end-user devices and local clusters rather than centralised data centres [282,320] is a reaction to the third Generation of IoT's *cloudification* [231]. At the same time, the trend away from cloudification is the foundational moment for the fourth Generation of IoT. Furthermore, sensors generally pose a trade-off between computational power, latency, cost, accuracy, energy consumption, and security [42,190,321,322]. To balance these trade-offs, additional computational power was introduced physically *closer*

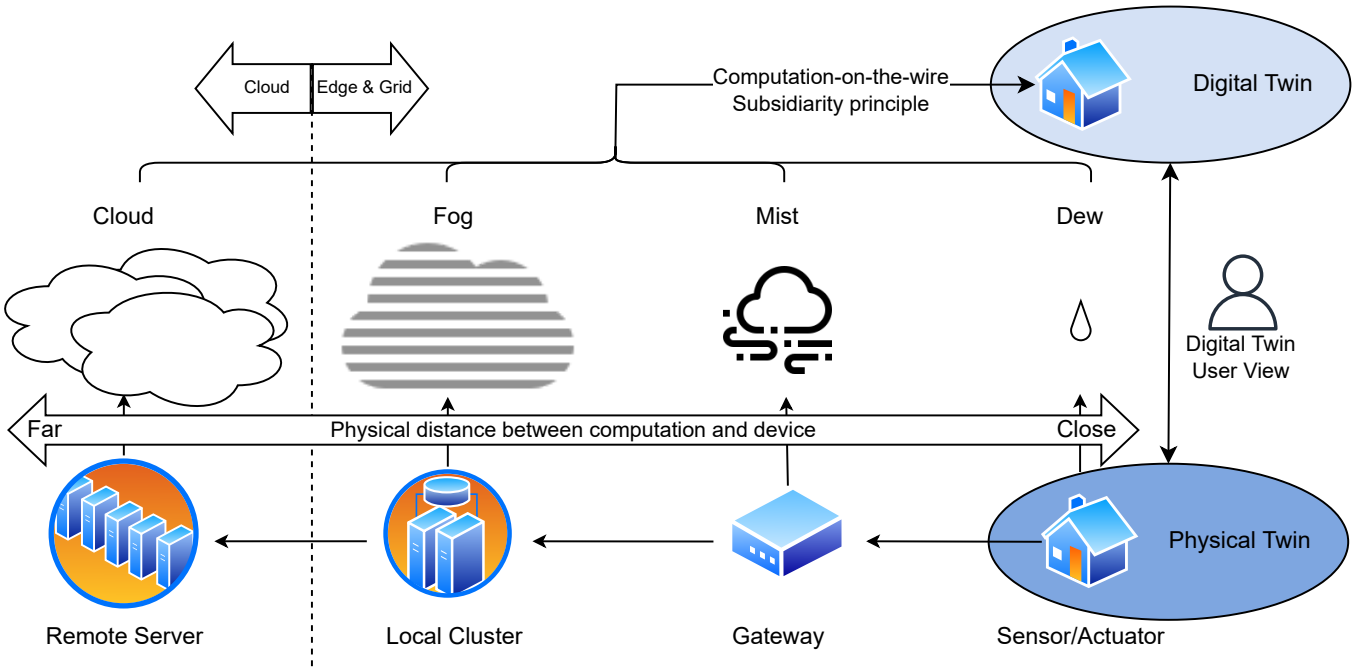

Figure 13. Computing Hydrology of Computation-on-the-wire. Computing power can be located locally at the sensor/actuator or more remotely at the gateway, a local cluster, or remote server. The computations form the basis for a DT. Computing in IoT and smart cities is often conceptualised with a hydrological terminology (i.e., cloud, fog, mist, and dew) that maps to the computing location and encapsulates different trade-offs in computational intensity, traffic, latency, accuracy, energy consumption, and security [42,190]. These trade-offs are best evaluated under the subsidiarity principle to decide where to place the computation physically. Edge or grid computing [149,150] is another terminology to covers all non-cloud computing. DTs rely heavily on these different data and computing sources while at the same time abstracting away their existence in the end users' view. Generated with http://draw.io.

an application layer mapping to the Virtual Environment. This architecture belongs to the earlier models and is focused more on the process of how to create basic IoT infrastructure. The perception layer describes the process of data acquisition usually provided from sensors, actuators, and tags. The middleware layer aggregates the data and provides APIs to access necessary information. When comparing the three-layer model to the two-layer model, middleware is the hidden layer that converts the real world's PT into its DT. However, most real systems stop short of complex analysis based on simulation and often limit itself to present raw data or aggregates.

The application layer offers useful services on top of the data. In a sense, those services are offered in the Virtual Environment of the DT (see Figure 2), but they heavily rely on the Analytical Environment to produce interesting services. Naming conventions for the three layers vary, but their basic functionality remains the same such as in ETSI M2M [164], SENSEI [364]. Additional layers always focus on very specific features and mostly subdivided the Application layer for Service-oriented Architecture [153], Resource-oriented Architecture [161] business applications [154], or Context Awareness [155].

5.2. The Analytical Environment of the Digital Twin - Models and Services

The Analytical Environment is often considered to be the DT itself [151] because it simulates the PTs behaviour. However, throughout this review paper, we have outlined more specific tasks for the Analytical Environment such as running simulations, producing aggregations, and providing services. We also have elaborated why a DT is more than just a simple simulation with real-world data. Nonetheless, it is the Analytical Environment that usually gives DTs and their Cognates the Smart label such as in Smart City.

The word *smart* generally allures to the superiority of a (connected, digitised) Thing compared to its dumb (unconnected, analogue) counterpart. Since connectivity and digitalisation on their own do not make anything *smart* but only provide accessibility, the smartness implies easy interaction with the Thing, new insights into the Thing, and new applications of the Thing. To attain any of these goals, models are required to represent reality, predict the future, and monitor ongoing processes. For DTs, this means that services [368,369] are provided on top of the data through the Analytical Environment to enrich a PT's functionality and simplify interactions. In general, most Things will require a DT to become smart.

However, the term smart has been miss-used. The previous description of *smart* is applicable to any meaningful service. The use of "*Smart*" as a prefix for services has proliferated to the point where it has become superfluous. Following the above argument, the prefix has been attached to anything that could have a DT or has implemented features that are indistinguishable from a DT. In the simplest case, a Smart Object provides Awareness, Representation, Interactivity, and Identity [12,257]. Smart Objects are concrete implementations of a DT (and even the FTs paradigm), but they limit the Thing to be a small-scale object with a fixed use. Smart Objects are also a form of data physicalisation [124] that provides the opportunity to interact with data beyond a visual medium.

As objects increase in size and become places, the terminology adapts to Smart Environment [320], Smart Home [370,371], Smart Office [371], Smart Museum [372], Smart City [9,15,48,198,199,253,323,370,373–375] and even Smart Regions [376] and Smart Communities [247,377]. Conceptually, these Smart Places [378] do not differ from Smart Objects [12] but only generalise it to sets of Things. This scaling aligns with the composability of DTs and allows for these concepts to be nested.

However, there is a difference to be made between physical Things and non-physical Things such as processes. The Thing in this context is a non-physical entity but interacts with physical entities in a defined way allowing to define smart processes such as Smart

Governance [379], Smart Manufacturing [380,381], Smart Construction [382], Smart Health Care Services [383], Smart Mobility [384], and Smart Experiences (for touristic activities)[385].

With everything labelled smart, it becomes more meaningful to talk about the kind of actions a Smart Thing can provide. Interestingly, it is these interactions that are also sought after in the immersive analytics context [68]. We locate actions on a scale of how independently they can react to external input:

1. Sensing: Only produce data
2. Reacting: Use threshold *on collected data* to trigger action
3. Interacting: Use *additional input* beyond collected data to change behaviour
4. Analysing: Use a *model to interpret collected data* when choosing actions
5. Proposing: Use a *decision model* to propose actions based on data models
6. Deciding: *Autonomously perform* all necessary action to interact with data.

The action space of Smart Things is constantly expanded and we cannot enumerate everything here. However, all of these developments can be understood as the Analytical Environment of a DT. Research has covered many approaches to provide the Analytical Environment needed to transform data into insights (see Table 5).

Table 5. Research Areas for Digital Twin Analytics

Domain	Description	Review
Ontologies	Make semantic sense of the data and produce knowledge and context	[155,224–226,386–388]
Middleware	Infrastructure to identify, process, store, secure and present information	[227–229,389,390]
Data preprocessing	Data aggregation and data fusion	[303,391–397]
Data mining	Cloud-based Big Data analysis on unstructured data	[398–405]
AI&ML	Machine learning and deep learning	[395,406–408]
Blockchain	Security, trust, and privacy through smart contracts	[409–413]
HCI	Human-Computer Interaction for improved access	[70,85,86,259,414–416]
ABM	Agent-based modelling Objects	[417–420]

5.3. The Virtual Environment of the Digital Twin - Access, Interaction and Representation through the Fused Twins Paradigm

The last environment we discuss provides the Human-Computer Interaction [421,422] with the DT. When implementing the FTs paradigm for a DT, the main focus lies on AR interactions with the DT. AR is only beginning to arrive in the world of SOs, the IoT, remote sensing, and DTs but has already been accepted as a crucial feature in the near future [423]. AR is of interest for FTs as it explicitly supports embodied interaction: interaction that is not agnostic to its physical and social context [74,80]. AR can also reduce negative split-attention effects [118,120], and appropriate visualisations of data can lower cognitive load [73] and enhance cognition [72]. The expectation is that AR will offer a new avenue to simplify interaction with SOs, the IoT, and DTs [3]. The FTs paradigm is a formalisation of the interaction between AR and DTs in a situated context.

Because the combination of technologies is only emergent now, most published material towards this end is either a prototype [3,101,363,424,425] or a proposal for future research directions [426–429]. The main reason is that applying AR on top of the already complex combination of SO, IoT, remote sensing, and DT stacks requires many software systems to run smoothly that are currently only being prototyped.

The Fusion of the the digital representation and the physical entity lies at the core of the FTs paradigm [3] but also of the more general Situated Analytics [5]. Interestingly, other research on the topic of Situated Analytics has often implemented FTs to some degree

but without the explicit mentioning of the DT aspect [363,424,425]. A notable exception is the 'Corsican Twins" [101] implementation of a FTs platform with an explicit DT powering the Situated Analytics. Short papers on future research directions in Situated Analytics clearly point towards FTs-like implementations [426–429]. We summarise ongoing and future research that is already covering a wide variety of topics (see Table 6). Some of the research explicitly mentioned the DT Approach or an Situated Analytics approach but often simply uses the available technology to demonstrate that their problem can be addressed [427,429,430], implicitly developing a DT and applying Situated Analytics.

Table 6. Recent and Ongoing Research Applying the Fused Twins Paradigm

		Environments				
Application Type	Year	Physical	Data	Analytical	Virtual	Connection
Infrastructure Management [431]	2009	*	✓	*	✓	-
City Information [22]	2009	*	✓	✓	✓	-
History [430]	2013	*	✓	-	✓	-
City Information (concept) [432]	2014	*	*	*	*	*
Shopping [433]	2015	*	✓	✓	*	-
IoT device interaction [424]	2016	✓	✓	-	✓	*
City Information [434]	2016	*	✓	*	✓	-
Shopping [435]	2018	*	✓	✓	*	-
Smart Campus [436]	2018	✓	✓	✓	✓	✓
Evacuation [437]	2018	*	-	-	✓	-
Medicine [438]	2018	✓	✓	-	✓	-
Facility Management [439]	2018	✓	✓	✓	✓	-
Rehabilitation [440]	2019	✓	✓	*	✓	-
IoT device visualisation [425]	2019	✓	-	-	✓	-
Seat Selection [441]	2020	*	✓	✓	*	-
Equipment Maintenance [442]	2020	*	✓	-	✓	-
In situ visualisation [101]	2020	*	✓	*	✓	*
Education [443]	2020	✓	✓	✓	✓	*
Training & Assistance [444]	2020	*	*	-	✓	*
Firefighting [429]	2021	-	-	✓	✓	-
General [3]	2021	✓	✓	✓	✓	*
IoT device interaction [363]	2021	✓	✓	✓	✓	-
General [100]	2022	✓	✓	✓	✓	*

Note: * Partially implemented or insufficient information

Since the FTs concept in particular and DT concept in general is not well defined yet, we analysed the literature for traces of it. To construct Table 6, we analysed the papers for the presence of implementations of any of the five components. For the Physical Environment,

simple and accessible example of FTs. Similarly, in the context of tourism, museums have created DTs of specific exhibits but could be expanded to included the entire exhibition to create an FTs. In the future, FTs could also be expended to communicate city planning initiatives to the public.

7. Conclusion

In this review paper, the relevant technologies and terminology behind the FTs concept for interacting with the DT of Smart Cities have been presented. Suitable examples of FTs applications across different literatures have been highlighted (see Table 6. Nonetheless, a stable and applicable out-of-the-box solution to fulfil the promised FTs for Smart Cities remains to be developed. DTs are also remain to be well-specified, which may cause delays in developing FTs. Here, we provided a thorough review of DTs and their cognates to address open points and a concrete definition that can be applied in retrospect to nearly all DTs. Compared to previous research that employed dashboards and other interfaces for Non-situated Analytics, FTs represent a special application of the Virtual Environment of a DT to provide Situated Analytics. The key technologies behind the FTs Paradigm for DTs are IoT, remote sensing, BIM/CIM, GIS, and Location services. The IoT, remote sensing, BIM/CIM, and GIS are used to assemble a DT and provide analytics, and Location services are used to place the Situated Analytics in the PT. Each of these technologies on its own is well-developed, but only now are they being combined into prototypes that can be classified as FTs. While many researchers have theorised about similar systems that could be built, few FTs full-fledged systems have been implemented. Given the plurality of fields that have converged on this general idea, we expect that the FTs paradigm will be a common theme in future technologies and may eventually be as common as a smartphone today.

Author Contributions: Conceptualisation, J.G.; methodology, J.G.; validation, J.G., T.T.; investigation, J.G., L.A., M.M., J.C.; resources, J.G.,C.H.,V.S.,R.S.; writing—original draft preparation, J.G.; writing—review and editing, J.G., T.T., L.A., J.C., M.M., V.S., R.S., C.H.; visualisation, J.G., M. M., J.C.; supervision, V.S., C.H., R.S; project administration, J.G., V.S., C.H.; funding acquisition, T.T., V.S., C.H., J.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by ETH Zurich grant number ETH-15 16-2 and J.G. was supported by an ETH Zurich Doc.Mobility Fellowship.

Institutional Review Board Statement: Not applicable

Informed Consent Statement: Not applicable

Acknowledgments: The authors thank Johanna Beyer and Zhu-Tian Chen for the feedback on the AR section. All diagrams were generated with templates from http://draw.io. And don't forget your towel.

Conflicts of Interest: The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

IoT	Internet of Things
DT	Digital Twin
PT	Physical Twin
FTs	Fused Twins
XR	Extended Reality
CPS	Cyber-Physical System
UC	Ubiquitous Computing
SO	Smart Object
WSN	Wireless Sensor Network
SN	Sensor Network
SAN	Sensor and Actuator Network
WSAN	Wireless Sensor and Actuator Networks
MR	Mixed Reality
AR	Augmented Reality
VR	Virtual Reality
AV	Augmented Virtuality
IaaS	Infrastructure as a Service
PaaS	Platform as a Service
SaaS	Software as a Service
SenaaS	Sensing as a Services
SemaaS	Semantics as a Service
RaaS	Robots as a Service
RSaaS	Remote Sensing as a Service
RFID	Radio Frequency Identification
OS	Operating System
ICT	Information and Communication Technologies
M2M	Machine-to-Machine
BAN	Body Area Networks
PAN	Personal Area Networks
LAN	Local Area Networks
WAN	Wide Area Networks
LPWAN	Low Power Wide Area Networks
BIM	Building Information Model
CIM	City Information Model
GIS	Geographic Information System
UAV	Unmanned Aerial Vehicles

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